

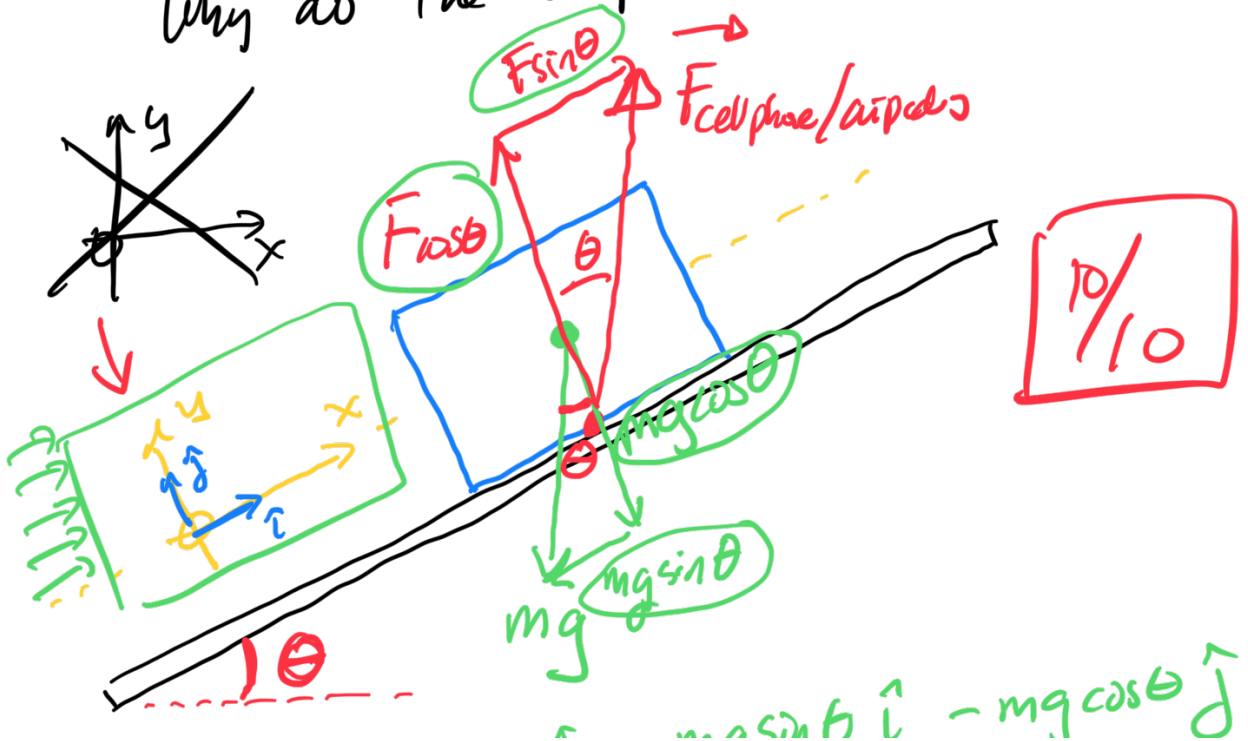
Physics 201 - Lecture 13

→ Assignment 4 (Forces)

Due next Friday, March 5th

→ Today: more about forces!
work through A4 problem

Why do the airports not move?



$-mg \hat{j}$

① Forces ② coord. sys. ③ break down

Step 1:

Free Body Diagram!

→ identify all of the forces
→ choose a coordinate system.

① gravity

② contact forces.

vector



$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \dots$$

$$-mg\hat{j} + \vec{F}_{\text{cell phone / airpods.}} = 0 \quad (\text{not moving})$$

$$\vec{F}_{\text{cell phase/carpals}} = +mg \hat{j}$$

Coordinate system → think about the expected motion

$$\vec{a} = a_x \hat{i} + \boxed{a_y} \hat{j}$$

$$a_y = 0 \quad !!$$

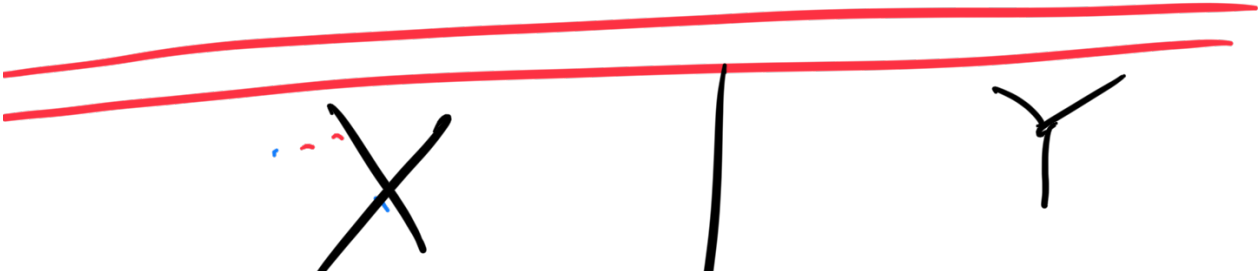
$$\vec{F}_{\text{net}} = \overset{\text{all motion}}{\boxed{F_{\text{net}}^x}} \hat{i} + \boxed{F_{\text{net}}^y} \hat{j}$$

$$= 0 \quad !!$$

$$= m \vec{a}$$

$$= m a_x \hat{i} + \boxed{m a_y} \hat{j}$$

$$= 0$$

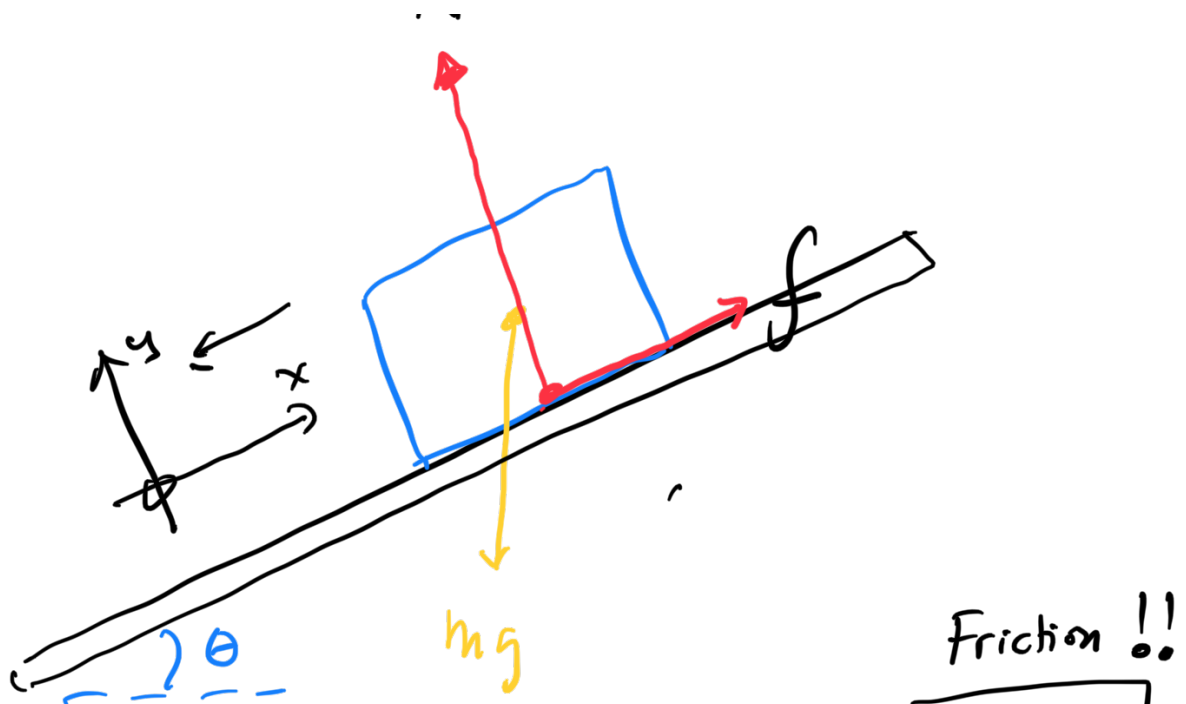


$\Sigma F_x = \max$ $\boxed{F \sin \theta - mg \sin \theta} = 0$ <i>diagram</i> $\downarrow$ $F \cancel{\sin \theta} = mg \cancel{\sin \theta}$ $\boxed{F = mg}$	$\Sigma F_y = \max$ $\boxed{F \cos \theta - mg \cos \theta} = 0$ <i>diagram.</i> $\downarrow$ $F \cancel{\cos \theta} = mg \cancel{\cos \theta}$ $\boxed{F = mg}$
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$$|H| \rightarrow |F|_{\text{cell phone/airports}} = mg$$

WHY?

why did it start moving at same angle?



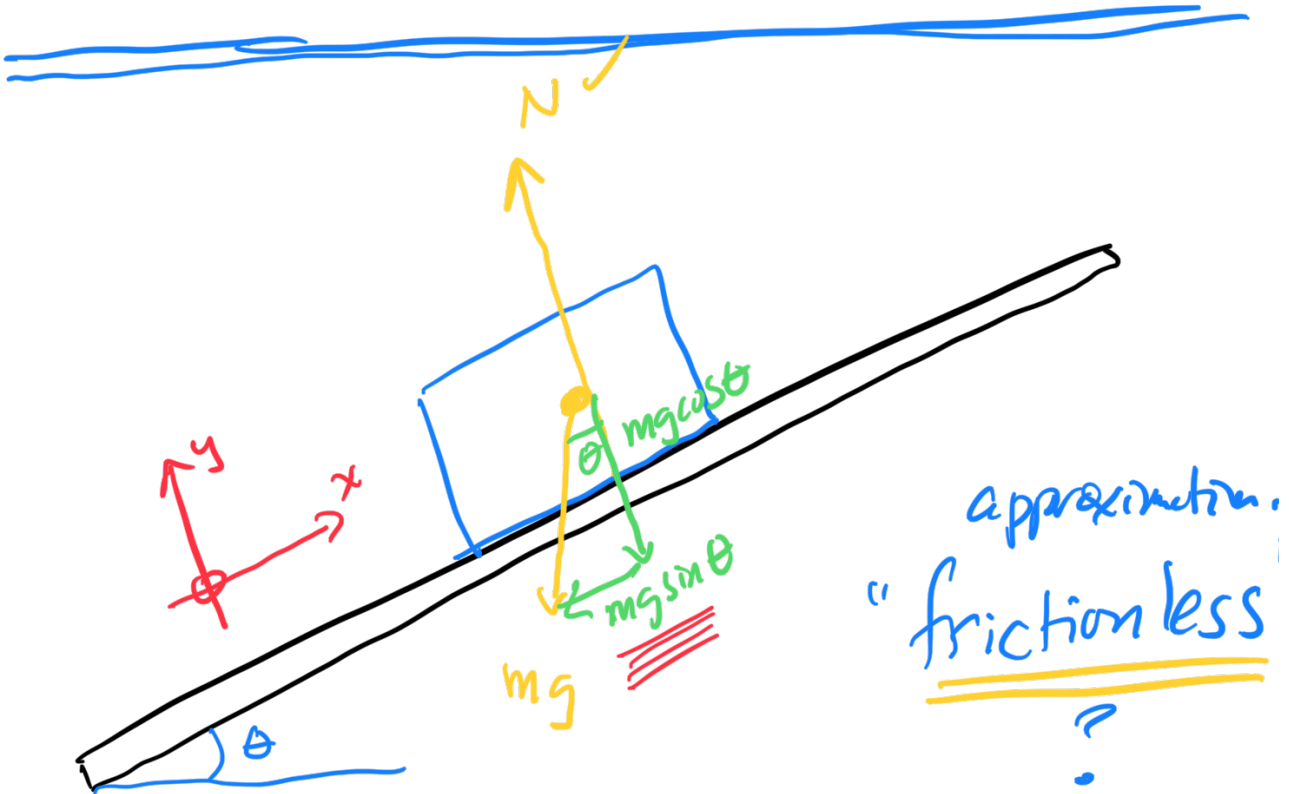
$$\boxed{\vec{F}_{\text{coll plane/air pads}}} = \boxed{\vec{F}_{\perp}} + \boxed{\vec{F}_{\parallel}}$$

Contact "Normal" Force.

$$\boxed{\vec{F}_{\text{contact}}} = \boxed{\vec{N}_{(\perp)}} + \boxed{\vec{f}_{(\parallel)}}$$

THINK about "opposes the motion"
 is expected $\rightarrow$

the motion.

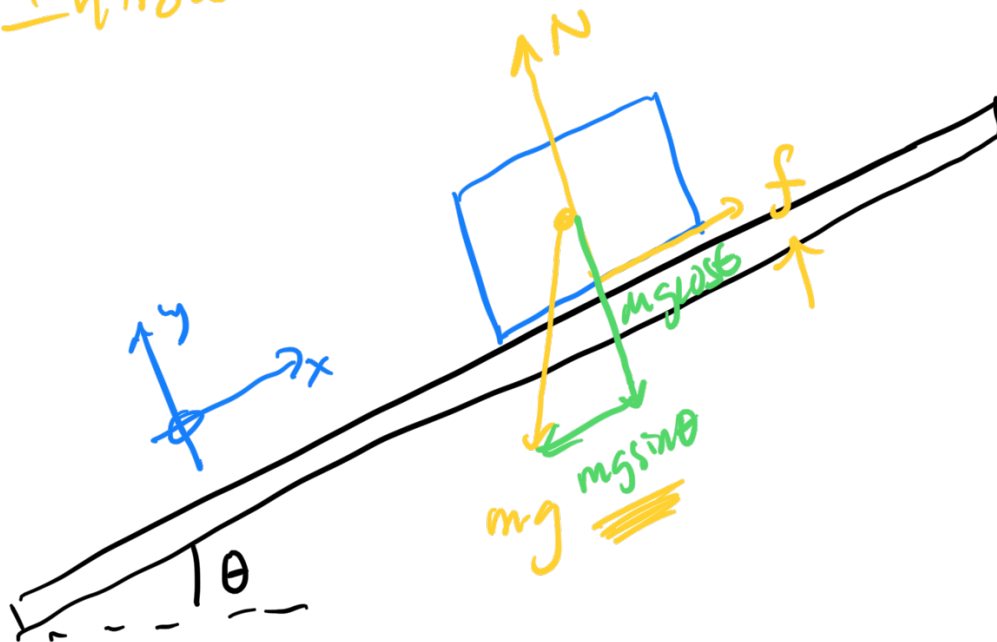


X	Y
$\sum F_x = ma_x$ $-mg \sin \theta = ma_x$	$\sum F_y = ma_y$ $N - mg \cos \theta = 0$
$a = a \sin \theta$	$N = mg \cos \theta$ $???$

$$a_x = -g \sin \theta$$

😊 I know the acceleration !!

Introduce friction!



$$\sum F_x = ma_x$$

$$\sum F_y = ma_y = 0$$

$$-mg \sin \theta + f = ma_x$$

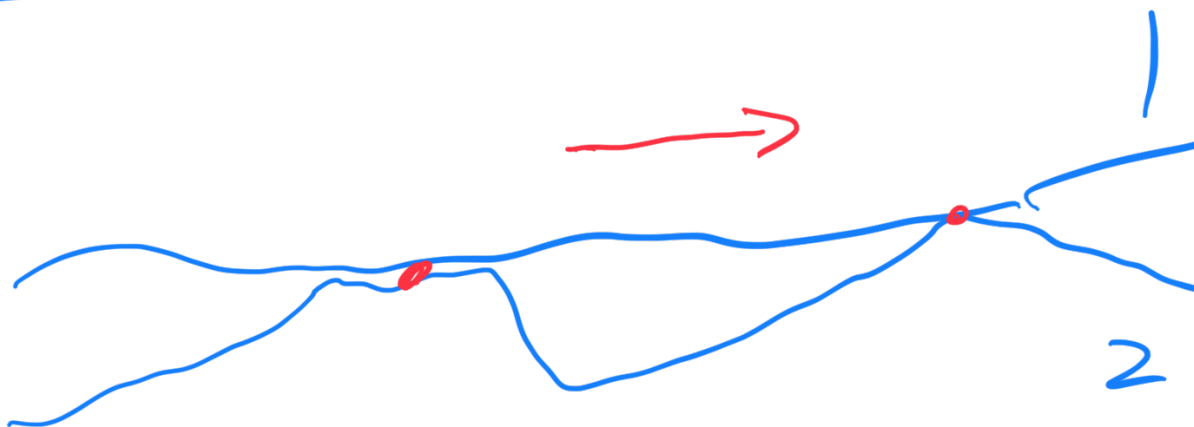
$$N - mg \cos \theta = 0$$

don't know f
don't know a_x

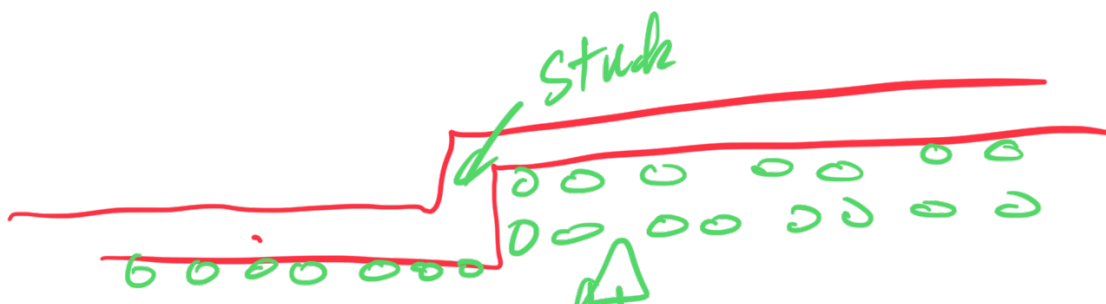
$$N = mg \cos \theta$$



Physics of friction



→ roughness
→ atoms, dislocation.



Summarize

$\mu \equiv$ coefficient of friction.
(experiment)

Small $\mu \rightarrow$ little friction (smooth, slippery)
large $\mu \rightarrow$ lots of friction (rough)

$\mu = 0 \rightarrow$ no friction!

$$\mu \equiv \frac{f}{N}$$

definition of friction.

$$f = \mu N$$



known from experiment.

$$-mg \sin \theta + \boxed{f} = ma_x$$

$$-mg \sin \theta + \mu \boxed{N} = ma_x$$

$mg \cos \theta$

~~$$-mg \sin \theta + \mu mg \cos \theta = ma_x$$~~

$$a_x = -g \sin \theta + \boxed{\mu g \cos \theta}$$

$\uparrow$
friction

more investigation